BASIC SKILLS BOOKLET

Algebra Basic Skills

Learn it. Practise it. Show you can do it.

Name: _____ Class: _____

How to use this booklet

1. **Learn.** Read the notes and the worked examples. Each example shows every step.

2. **Practise.** Do **Practice A**, then **Practice B**. Check your answers at the back of the booklet and fix any mistakes.

3. **Skill Check.** When you're ready, do the Skill Check **on your own, without notes**. Your teacher will mark it.

My progress

	Skill	Practice A done	Practice B done	Skill Check (teacher)
1	**Collecting Like Terms** I can add and subtract like terms, e.g. $3a + 4a = 7a$	☐	☐	/4
2	**Multiplying Algebra** I can multiply terms, e.g. $2a \times 3b = 6ab$ and $x \times x = x^2$	☐	☐	/4
3	**Dividing Algebra** I can divide terms, e.g. $8x \div 2 = 4x$ and $6ab \div 2a = 3b$	☐	☐	/4
4	**Expanding Brackets** I can expand brackets, e.g. $3(x+2) = 3x+6$ and $-2(x-4) = -2x + 8$	☐	☐	/4
5	**Factorising into Brackets** I can factorise, e.g. $2x + 6 = 2(x + 3)$	☐	☐	/4
	Mixed Review (extra practice)	☐		**Total** /20

The answers to the **practice** questions are at the back. The Skill Checks have no answers in this booklet — that's where you show what you can do!

1 Collecting Like Terms

LEARN

Goal: I can add and subtract like terms, e.g. $3a + 4a = 7a$

Like terms have **exactly the same letter part**. You can only add or subtract like terms.

Like terms ✓	NOT like terms ✗
$3a$ and $5a$	$3a$ and $5b$ (different letters)
x and $7x$	x and 7 (one has no letter)
$2x^2$ and x^2	x^2 and x (x^2 is not the same as x)
$4ab$ and ab	ab and a

Think of the letters as objects. 3 apples + 4 apples = 7 apples, so $3a + 4a = 7a$. But 3 apples + 2 bananas can't be joined, so $3a + 2b$ stays as $3a + 2b$.

A letter on its own means 1 of it. a is the same as $1a$. So $a + a + a = 3a$.

Worked examples

$3a + 4a$

3 lots of a plus 4 lots of a

$= \mathbf{7a}$

$6x - 2x$

6 take away 2 is 4

$= \mathbf{4x}$

$5y - y$

y means $1y$, and $5 - 1 = 4$

$= \mathbf{4y}$

$2a + 3b + 4a + b$

The a's: $2a + 4a = 6a$

The b's: $3b + b = 4b$

$= \mathbf{6a + 4b}$

$x^2 + 3x + 2x^2$

The x^2's: $x^2 + 2x^2 = 3x^2$

The x's: just $3x$

$= \mathbf{3x^2 + 3x}$

$7m + 2 - 3m$

The m's: $7m - 3m = 4m$

The numbers: just 2

$= \mathbf{4m + 2}$

Remember: The sign ($+$ or $-$) in **front** of a term belongs to that term. Circle or highlight each type of term in a different colour before you add them up.

1 Collecting Like Terms

PRACTICE A

Simplify by collecting like terms. Show your working in each box.

1 $a + a$

$=$ _____

2 $a + a + a$

$=$ _____

3 $2a + 3a$

$=$ _____

4 $4x + 5x$

$=$ _____

5 $3m + m$

$=$ _____

6 $7b - 3b$

$=$ _____

7 $6y - y$

$=$ _____

8 $10x - 4x$

$=$ _____

9 $2p + 3p + p$

$=$ _____

10 $8c - 2c$

$=$ _____

11 $x^2 + x^2$

$=$ _____

12 $3x^2 + 4x^2$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong before moving on.**

1 Collecting Like Terms

PRACTICE B

Simplify by collecting like terms. Some have two types of term. Show your working in each box.

1 $2a + 3b + 4a$

$=$ _____

2 $5x + 2y + x$

$=$ _____

3 $3a + 2b + a + 4b$

$=$ _____

4 $4x + 3 + 2x$

$=$ _____

5 $7m - 2m + 3n$

$=$ _____

6 $5a + 2b - 3a$

$=$ _____

7 $6x + 4 - 2x + 1$

$=$ _____

8 $8x - 3x + 2y$

$=$ _____

9 $4ab + 2ab$

$=$ _____

10 $x^2 + 2x + 3x$

$=$ _____

11 $2x^2 + x + x^2$

$=$ _____

12 $3y^2 + 2y + y^2 + 4y$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong before moving on.**

1 Collecting Like Terms

SKILL CHECK

Name: _____

Teacher mark: ____ / 4

Simplify by collecting like terms. Do this on your own — no notes. Show your working, and write your final answer on the line.

1 $5a + 2a$

$=$ _____

2 $8x - 3x$

$=$ _____

3 $4a + 3b + 2a$

$=$ _____

4 $x^2 + 5x + 2x^2$

$=$ _____

2 Multiplying Algebra

LEARN

Goal: I can multiply terms, e.g. $2a \times 3b = 6ab$ and $x \times x = x^2$

In algebra we **don't write the $\times$ sign**. Letters and numbers that are multiplied are written next to each other, **number first**.

We write...	...it means
$3a$	$3 \times a$
ab	$a \times b$
$5xy$	$5 \times x \times y$
x^2	$x \times x$ (say "x squared")

How to multiply terms

1. **Multiply the numbers** together.
2. **Write the letters** after the number, in alphabetical order.
3. If the **same letter** appears twice, write it as a **square**: $x \times x = x^2$.

Worked examples

$4 \times a$

Number first, then the letter

$= \mathbf{4a}$

$a \times b$

Write the letters together

$= \mathbf{ab}$

$2a \times 4$

Numbers: $2 \times 4 = 8$

Letters: a

$= \mathbf{8a}$

$2a \times 3b$

Numbers: $2 \times 3 = 6$

Letters: a and b

$= \mathbf{6ab}$

$x \times x$

The same letter twice is "squared"

$= \mathbf{x^2}$

$3x \times 2x$

Numbers: $3 \times 2 = 6$

Letters: $x \times x = x^2$

$= \mathbf{6x^2}$

Remember: When you **multiply**, the numbers multiply ($3 \times 2 = 6$) — don't add them! And $x \times x$ is x^2, not $2x$.

Going further: $x \times x \times x = x^3$ ("x cubed"), and $x^2 \times x = x^3$.

2 Multiplying Algebra

PRACTICE A

Simplify. Show your working in each box.

1 $4 \times a$

$=$ _____

2 $5 \times x$

$=$ _____

3 $a \times b$

$=$ _____

4 $x \times y$

$=$ _____

5 $2 \times 3a$

$=$ _____

6 $3x \times 4$

$=$ _____

7 $3 \times 5m$

$=$ _____

8 $6 \times 2x$

$=$ _____

9 $a \times a$

$=$ _____

10 $y \times y$

$=$ _____

11 $2a \times b$

$=$ _____

12 $p \times q \times r$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong before moving on.**

2 Multiplying Algebra

PRACTICE B

Simplify. Watch out for the same letter twice! Show your working in each box.

1 $2a \times 3b$

$=$ _____

2 $4x \times 2y$

$=$ _____

3 $5a \times 2b$

$=$ _____

4 $2ab \times 3$

$=$ _____

5 $3a \times a$

$=$ _____

6 $2x \times 5x$

$=$ _____

7 $4m \times 3m$

$=$ _____

8 $3x \times 2x$

$=$ _____

9 $a \times 2b \times 3$

$=$ _____

10 $x \times x \times x$

$=$ _____

11 $x^2 \times x$

$=$ _____

12 $2x^2 \times 3x$

$=$ _____

✓ Check your answers at the back. Fix any you got wrong before moving on.

2 Multiplying Algebra

SKILL CHECK

Name: _____

Teacher mark: ____ / 4

Simplify. Do this on your own — no notes. Show your working, and write your final answer on the line.

1 $3 \times 5a$

$=$ _____

2 $2a \times 4b$

$=$ _____

3 $x \times x$

$=$ _____

4 $3x \times 2x$

$=$ _____

3 Dividing Algebra

LEARN

Goal: I can divide terms, e.g. $8x \div 2 = 4x$ and $6ab \div 2a = 3b$

Dividing is the opposite of multiplying. A division can also be written as a fraction: $6a \div 2$ is the same as $\frac{6a}{2}$.

How to divide terms

1. **Divide the numbers.**
2. **Cancel any letter** that is on the top **and** the bottom.
3. Write down what is left.

Why can letters cancel?. Anything divided by itself is 1. Just like $5 \div 5 = 1$, $a \div a = 1$. So $\frac{ab}{a} = b$.

Worked examples

$6a \div 2$

Numbers: $6 \div 2 = 3$

Letters: a stays

$= \mathbf{3a}$

$ab \div a$

The a cancels

b is left

$= \mathbf{b}$

$8x \div x$

The x cancels

8 is left

$= \mathbf{8}$

$12ab \div 4a$

Numbers: $12 \div 4 = 3$

a cancels, b is left

$= \mathbf{3b}$

$x^2 \div x$

x^2 is $x \times x$

One x cancels, one is left

$= \mathbf{x}$

$6x^2 \div 3x$

Numbers: $6 \div 3 = 2$

Letters: $x^2 \div x = x$

$= \mathbf{2x}$

Remember: If **all** the letters cancel, you are left with just a number, e.g. $5a \div a = 5$. Write x^2 as $x \times x$ if it helps you see what cancels.

3 Dividing Algebra

PRACTICE A

Simplify. Show your working in each box.

1 $8a \div 2$

$=$ _____

2 $12x \div 3$

$=$ _____

3 $10b \div 5$

$=$ _____

4 $20y \div 4$

$=$ _____

5 $9x \div 3$

$=$ _____

6 $15p \div 5$

$=$ _____

7 $18x \div 9$

$=$ _____

8 $6m \div 6$

$=$ _____

9 $ab \div b$

$=$ _____

10 $xy \div x$

$=$ _____

11 $5a \div a$

$=$ _____

12 $7x \div x$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong before moving on.**

3 Dividing Algebra

PRACTICE B

Simplify. Cancel the letters that are on the top and the bottom. Show your working in each box.

1 $6ab \div 2a$

$=$ _____

2 $12xy \div 4y$

$=$ _____

3 $10ab \div 5b$

$=$ _____

4 $8xy \div 2x$

$=$ _____

5 $14ab \div 7$

$=$ _____

6 $x^2 \div x$

$=$ _____

7 $a^2 \div a$

$=$ _____

8 $4x^2 \div x$

$=$ _____

9 $6x^2 \div 2x$

$=$ _____

10 $15a^2 \div 5a$

$=$ _____

11 $20x^2 \div 4x$

$=$ _____

12 $9x^2 \div 3x$

$=$ _____

✓ Check your answers at the back. Fix any you got wrong before moving on.

3 Dividing Algebra

SKILL CHECK

Name: _____

Teacher mark: ____ / 4

Simplify. Do this on your own — no notes. Show your working, and write your final answer on the line.

1 $12x \div 4$

$=$ _____

2 $ab \div b$

$=$ _____

3 $10ab \div 5a$

$=$ _____

4 $8x^2 \div 2x$

$=$ _____

4 Expanding Brackets

LEARN

Goal: I can expand brackets, e.g. $3(x+2) = 3x+6$ and $-2(x-4) = -2x+8$

Expanding means getting rid of the brackets by multiplying. A number or letter right in front of a bracket means **multiply everything inside** the bracket by it.

Multiply the outside by EVERY term inside
Draw arrows so you don't miss one:

$$3(x+2) \longrightarrow 3\times x \text{ and } 3\times 2 \longrightarrow 3x+6$$

Worked examples

$4(a+5)$

$4\times a = 4a$
$4\times 5 = 20$
$= \mathbf{4a+20}$

$2(x-3)$

$2\times x = 2x$
$2\times(-3) = -6$
$= \mathbf{2x-6}$

$x(x+5)$

$x\times x = x^2$
$x\times 5 = 5x$
$= \mathbf{x^2+5x}$

Negative numbers outside the bracket

When the number outside is **negative**, use the sign rules:

Signs	Answer
same signs: $+\times+$ or $-\times-$	**positive** $(+)$
different signs: $+\times-$ or $-\times+$	**negative** $(-)$

$-2(x+3)$

$-2\times x = -2x$
$-2\times 3 = -6$
$= \mathbf{-2x-6}$

$-3(x-4)$

$-3\times x = -3x$
$-3\times(-4) = +12$
$= \mathbf{-3x+12}$

$-(x-5)$

A minus on its own means -1
$-1\times x = -x,\ -1\times(-5) = +5$
$= \mathbf{-x+5}$

Remember: The outside term multiplies **every** term inside: $3(x+2)$ is $3x+6$, **not** $3x+2$. With a negative outside, **every sign inside changes**: $-2(x-4) = -2x+8$.

4 Expanding Brackets

PRACTICE A

Expand. Show your working in each box.

1 $2(x+3)$

$=$ _____

2 $3(a+4)$

$=$ _____

3 $5(x+1)$

$=$ _____

4 $6(x+2)$

$=$ _____

5 $4(y-2)$

$=$ _____

6 $2(m-5)$

$=$ _____

7 $3(2x+1)$

$=$ _____

8 $2(4a+3)$

$=$ _____

9 $5(2y-1)$

$=$ _____

10 $x(x+2)$

$=$ _____

11 $a(a+5)$

$=$ _____

12 $x(x-3)$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong before moving on.**

4 Expanding Brackets

PRACTICE B

Expand. Be careful with the negative signs! Show your working in each box.

1 $-2(x+3)$

$=$ _____

2 $-3(a+4)$

$=$ _____

3 $-4(x-2)$

$=$ _____

4 $-5(y-1)$

$=$ _____

5 $-(x+7)$

$=$ _____

6 $-(a-3)$

$=$ _____

7 $2(3-x)$

$=$ _____

8 $4(2y-3)$

$=$ _____

9 $-2(3x-1)$

$=$ _____

10 $x(x-6)$

$=$ _____

11 $-x(x+2)$

$=$ _____

12 $-a(a-4)$

$=$ _____

✓ Check your answers at the back. Fix any you got wrong before moving on.

4 Expanding Brackets

SKILL CHECK

Name: _____

Teacher mark: ____ / 4

Expand. Do this on your own — no notes. Show your working, and write your final answer on the line.

1 $3(x+5)$ ☐

$=$ _____

2 $2(4a-3)$ ☐

$=$ _____

3 $x(x-2)$ ☐

$=$ _____

4 $-4(x-3)$ ☐

$=$ _____

5 Factorising into Brackets

LEARN

Goal: I can factorise, e.g. $2x + 6 = 2(x + 3)$

Factorising is the **opposite of expanding**. We put the brackets back in.

$$2(x+3) \xrightarrow{\text{expand}} 2x+6 \qquad 2x+6 \xrightarrow{\text{factorise}} 2(x+3)$$

How to factorise

1. Find the **biggest number** (and/or **letter**) that goes into **every** term. Write it outside the bracket.
2. **Divide** each term by it. Write the answers inside the bracket.
3. **Check** by expanding — you should get back to the question.

Worked examples

$5a + 10$

5 goes into both

$5a \div 5 = a,\ 10 \div 5 = 2$

$= \mathbf{5(a+2)}$

$6x - 9$

3 goes into both

$6x \div 3 = 2x,\ -9 \div 3 = -3$

$= \mathbf{3(2x-3)}$

$x^2 + 4x$

x goes into both

$x^2 \div x = x,\ 4x \div x = 4$

$= \mathbf{x(x+4)}$

$3x^2 + 6x$

$3x$ goes into both

$3x^2 \div 3x = x,\ 6x \div 3x = 2$

$= \mathbf{3x(x+2)}$

$14 - 7x$

7 goes into both

$14 \div 7 = 2,\ -7x \div 7 = -x$

$= \mathbf{7(2-x)}$

$8x - 12$

4 is the **biggest**

(2 works, but isn't the biggest)

$= \mathbf{4(2x-3)}$

Remember: Keep the sign with each term: in $6x - 9$ the second term is -9, so the bracket has a minus: $3(2x - 3)$. Always **check by expanding** your answer.

5 Factorising into Brackets

PRACTICE A

Factorise. Show your working in each box.

1 $3x + 6$

$=$ _____

2 $2a + 8$

$=$ _____

3 $5x + 15$

$=$ _____

4 $7m + 7$

$=$ _____

5 $4y - 12$

$=$ _____

6 $2x - 10$

$=$ _____

7 $6a + 9$

$=$ _____

8 $10x + 15$

$=$ _____

9 $x^2 + 3x$

$=$ _____

10 $a^2 + 5a$

$=$ _____

11 $x^2 - 2x$

$=$ _____

12 $y^2 + y$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong before moving on.**

5 Factorising into Brackets

PRACTICE B

Factorise fully. Take out the biggest factor you can. Show your working in each box.

1 $8x - 12$

$=$ _____

2 $6a - 15$

$=$ _____

3 $12y + 18$

$=$ _____

4 $20x - 10$

$=$ _____

5 $9m - 6$

$=$ _____

6 $14 - 7x$

$=$ _____

7 $x^2 - 7x$

$=$ _____

8 $ab + 3a$

$=$ _____

9 $3x^2 + 6x$

$=$ _____

10 $2a^2 - 8a$

$=$ _____

11 $5y^2 + 10y$

$=$ _____

12 $4x^2 - 6x$

$=$ _____

✓ **Check your answers at the back. Fix any you got wrong before moving on.**

5 Factorising into Brackets

SKILL CHECK

Name: _____

Teacher mark: ____ / 4

Factorise fully. Do this on your own — no notes. Show your working, and write your final answer on the line.

1 $5x + 20$

$=$ _____

2 $6a - 9$

$=$ _____

3 $x^2 + 8x$

$=$ _____

4 $4y^2 - 12y$

$=$ _____

★ Mixed Review

EXTRA PRACTICE

A mix of all five skills. Look carefully — which skill do you need? Answers are at the back.

1 $5a + 2a$

$=$ _____

2 $3 \times 2x$

$=$ _____

3 $12y \div 3$

$=$ _____

4 $2(x + 5)$

$=$ _____

5 $3x + 12$

$=$ _____

6 $4x + 3y + 2x$

$=$ _____

7 $a \times 4b$

$=$ _____

8 $10ab \div 2b$

$=$ _____

9 $x(x + 1)$

$=$ _____

10 $x^2 + 7x$

$=$ _____

11 $8m - 3m$

$=$ _____

12 $5x \times x$

$=$ _____

13 $6x^2 \div x$

$=$ _____

14 $4(2a - 1)$

$=$ _____

15 $5y - 20$

$=$ _____

16 $2x^2 + 3x + x^2$

$=$ _____

17 $4a \times 2a$

$=$ _____

18 $16xy \div 8x$

$=$ _____

19 $-3(x - 2)$

$=$ _____

20 $6x^2 + 9x$

$=$ _____

Answers to the Practice Questions

Terms can be in a different order and still be right — e.g. $2b + 3a$ is the same as $3a + 2b$.

1. Collecting Like Terms

Practice A

1. $2a$
2. $3a$
3. $5a$
4. $9x$
5. $4m$
6. $4b$
7. $5y$
8. $6x$
9. $6p$
10. $6c$
11. $2x^2$
12. $7x^2$

Practice B

1. $6a + 3b$
2. $6x + 2y$
3. $4a + 6b$
4. $6x + 3$
5. $5m + 3n$
6. $2a + 2b$
7. $4x + 5$
8. $5x + 2y$
9. $6ab$
10. $x^2 + 5x$
11. $3x^2 + x$
12. $4y^2 + 6y$

2. Multiplying Algebra

Practice A

1. $4a$
2. $5x$
3. ab
4. xy
5. $6a$
6. $12x$
7. $15m$
8. $12x$
9. a^2
10. y^2
11. $2ab$
12. pqr

Practice B

1. $6ab$
2. $8xy$
3. $10ab$
4. $6ab$
5. $3a^2$
6. $10x^2$
7. $12m^2$
8. $6x^2$
9. $6ab$
10. x^3
11. x^3
12. $6x^3$

3. Dividing Algebra

Practice A

1. $4a$
2. $4x$
3. $2b$
4. $5y$
5. $3x$
6. $3p$
7. $2x$
8. m
9. a
10. y
11. 5
12. 7

Practice B

1. $3b$
2. $3x$
3. $2a$
4. $4y$
5. $2ab$
6. x
7. a
8. $4x$
9. $3x$
10. $3a$
11. $5x$
12. $3x$

4. Expanding Brackets

Practice A

1. $2x + 6$
2. $3a + 12$
3. $5x + 5$
4. $6x + 12$
5. $4y - 8$
6. $2m - 10$
7. $6x + 3$
8. $8a + 6$
9. $10y - 5$
10. $x^2 + 2x$
11. $a^2 + 5a$
12. $x^2 - 3x$

Practice B

1. $-2x - 6$
2. $-3a - 12$
3. $-4x + 8$
4. $-5y + 5$
5. $-x - 7$
6. $-a + 3$
7. $6 - 2x$
8. $8y - 12$
9. $-6x + 2$
10. $x^2 - 6x$
11. $-x^2 - 2x$
12. $-a^2 + 4a$

5. Factorising into Brackets

Practice A

1. $3(x + 2)$
2. $2(a + 4)$
3. $5(x + 3)$
4. $7(m + 1)$
5. $4(y - 3)$
6. $2(x - 5)$
7. $3(2a + 3)$
8. $5(2x + 3)$
9. $x(x + 3)$
10. $a(a + 5)$
11. $x(x - 2)$
12. $y(y + 1)$

Practice B

1. $4(2x - 3)$
2. $3(2a - 5)$
3. $6(2y + 3)$
4. $10(2x - 1)$
5. $3(3m - 2)$
6. $7(2 - x)$
7. $x(x - 7)$
8. $a(b + 3)$
9. $3x(x + 2)$
10. $2a(a - 4)$
11. $5y(y + 2)$
12. $2x(2x - 3)$

⋆ Mixed Review

1. $7a$
2. $6x$
3. $4y$
4. $2x + 10$
5. $3(x + 4)$
6. $6x + 3y$
7. $4ab$
8. $5a$
9. $x^2 + x$
10. $x(x + 7)$
11. $5m$
12. $5x^2$
13. $6x$
14. $8a - 4$
15. $5(y - 4)$
16. $3x^2 + 3x$
17. $8a^2$
18. $2y$
19. $-3x + 6$
20. $3x(2x + 3)$